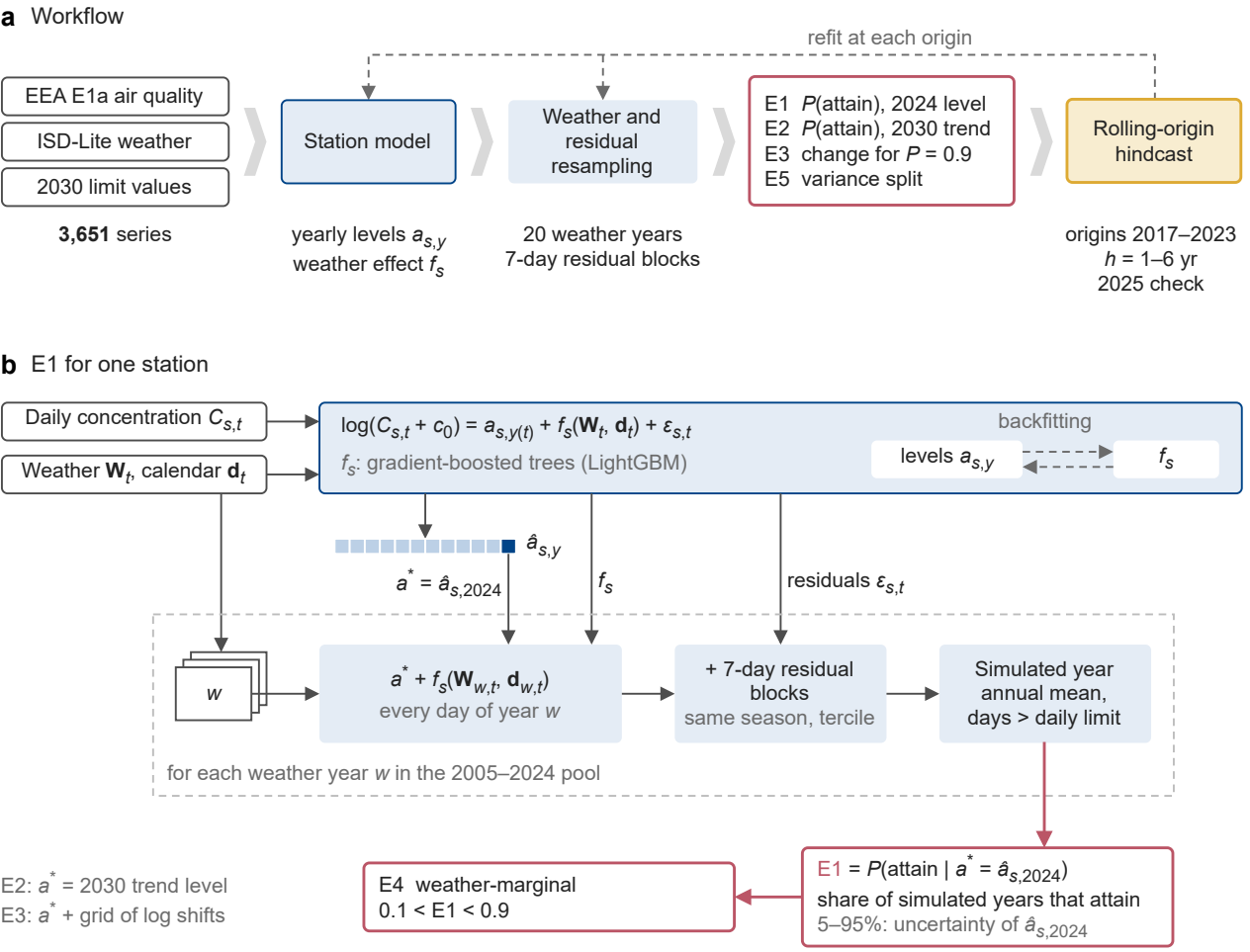

a Workflow
refit at each origin
EEA E1a air quality
ISD-Lite weather
2030 limit values
3,651 series
Station model
yearly levels $a_{s,y}$
weather effect f_s
Weather and residual resampling
20 weather years
7-day residual blocks
E1 P(attain), 2024 level
E2 P(attain), 2030 trend
E3 change for P = 0.9
E5 variance split
Rolling-origin hindcast
origins 2017–2023
h = 1–6 yr
2025 check
b E1 for one station
Daily concentration $C_{s,t}$
Weather $\mathbf{W}_t$, calendar $\mathbf{d}_t$
$\log(C_{s,t} + c_0) = a_{s,y(t)} + f_s(\mathbf{W}_t, \mathbf{d}_t) + \varepsilon_{s,t}$
f_s: gradient-boosted trees (LightGBM)
backfitting
levels $a_{s,y}$
f_s
$\hat{a}_{s,y}$
$a^* = \hat{a}_{s,2024}$
f_s
residuals $\varepsilon_{s,t}$
w
$a^* + f_s(\mathbf{W}_{w,t}, \mathbf{d}_{w,t})$
every day of year w
+ 7-day residual blocks
same season, tercile
Simulated year annual mean, days > daily limit
for each weather year w in the 2005–2024 pool
E1 = $P(\text{attain} \mid a^* = \hat{a}_{s,2024})$
share of simulated years that attain
5–95%: uncertainty of $\hat{a}_{s,2024}$
E4 weather-marginal
0.1 < E1 < 0.9
E2: a^* = 2030 trend level
E3: a^* + grid of log shifts